Network Access and Media Guest Controls

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a network access system for temporary conference media accounts. Media guests may access the press upload service, and the system must block their access to the organizer network. Network administrators can revoke media guest access. Leave the session duration open for review.

1.3 Definitions, acronyms and abbreviations

No project-specific terms were supplied.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a network access system for temporary conference media accounts.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- allow media guests to access the press upload service and block their access to the organizer network - FR-001
- allow network administrators to revoke media guest access - FR-002

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
User	guests	Uses permitted functions; each source class keeps its specified permissions.
Admin	Network administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Leave the session duration open for review.
- How are the stated roles and authorized permissions assigned?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.4 Communications interfaces

Related requirements: FR-001, FR-002. Interface details beyond those requirements remain unspecified.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall allow media guests to access the press upload service and block their access to the organizer network.

Traceability: E-002, TEST-001, TEST-002

FR-002 · Must · explicit

The system shall allow network administrators to revoke media guest access.

Traceability: E-003, TEST-003, TEST-004, TEST-005

3.3 Performance requirements

No separate requirement was specified in the supplied brief.

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

No separate requirement was specified in the supplied brief.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

No separate requirement was specified in the supplied brief.

3.6.4 Maintainability

No separate requirement was specified in the supplied brief.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: Source-defined permissions and restricted resources

Scenario: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Media guests may access the press upload service, and the system must block their access to the organizer network.

Hypothetical actor or cause: Unconfirmed actor or cause (planning hypothesis)

Potential consequence: Unauthorized access or modification outside the stated permission boundary could occur.

Linked requirements: FR-001

Original evidence: E-002

Residual exposure: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

A.2 THR-002: Access grants and revocations

Scenario: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A formerly authorized principal could continue using access that is no longer approved.

Linked requirements: FR-002

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

- Decision Q-b0e81d2ce472 is TBD.

B.1 RISK-001

Derived risk affecting Source-defined permissions and restricted resources: Unauthorized access or modification outside the stated permission boundary could occur.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Media guests may access the press upload service, and the system must block their access to the organizer network.

Proposed treatment: Review the failure or abuse path against the linked source obligations. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-001

Original evidence: E-002

B.2 RISK-002

Derived risk affecting Access grants and revocations: A formerly authorized principal could continue using access that is no longer approved.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Proposed treatment: Trace permission changes through subsequent access decisions and any active access; confirm revocation timing and session semantics before baselining them. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002

Original evidence: E-003

B.3 RISK-003

An unresolved source decision may delay validation: Leave the session duration open for review.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: The source leaves this decision unresolved; no delivery or technical conclusion is inferred.

Proposed treatment: Obtain a stakeholder decision before baselining affected behavior; retain the unknown and revalidate resulting changes.

Linked requirements:

Original evidence: E-004

C SDLC recommendation and delivery workflow

Proposed lifecycle: DevSecOps

The DevSecOps model is selected based on derived evidence (E-001) that security decisions benefit from integrated security review and testing throughout the delivery lifecycle. This factor provides a 12-point advantage over other models, with Hybrid (+8 points) and Spiral (+5 points) receiving lower scores due to insufficient differentiation in their alignment with continuous security assurance. The score of 52 outperforms all alternatives, establishing DevSecOps as the most suitable model under current evidence. However, unknown factors such as budget, delivery constraints, failure consequences, feedback loops, legacy systems, regulation, schedule, stability, uncertainty, and verification are explicitly acknowledged and not assumed—confirming team capacity and operational realities is required before adoption. The existing requirements FR-001 and FR-002 relate to access control for media guests and revocation by administrators, which align with the need for continuous security validation in DevSecOps. No other model demonstrates a distinct advantage in derived security assurance or integration of testing into delivery phases.

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- security_assurance = integrated security review (derived). Security decisions benefit from security review and tests throughout delivery. Original evidence: E-001.

Candidate	Suitability points
DevSecOps	52
Hybrid	48
Spiral	45
Agile	40
V-Model	40
Waterfall	40
Incremental	40

C.1.1 DevSecOps suitability reasons

- Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-001)

C.1.2 Hybrid suitability reasons

- Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-001)

C.1.3 Spiral suitability reasons

- Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-001)

C.1.4 Agile suitability reasons

- No differentiating input evidence

C.1.5 V-Model suitability reasons

- No differentiating input evidence

C.1.6 Waterfall suitability reasons

- No differentiating input evidence

C.1.7 Incremental suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall allow media guests to access the press upload service and block their access to the organizer network.; FR-002: The system shall allow network administrators to revoke media guest access.
- Activity: Record unresolved source decision: Leave the session duration open for review.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.2 Design and test design

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Design the implementation for the source-defined work group: FR-001: The system shall allow media guests to access the press upload service and block their access to the organizer network.; FR-002: The system shall allow network administrators to revoke media guest access.
- Activity: Refine the early unit, integration, system and acceptance test plan without choosing unresolved product technologies.
- Deliverable: Proposed design linked to canonical requirements
- Deliverable: Unit, integration, system and acceptance test plan
- Entry criterion: The requirements baseline and early test plan have been reviewed.
- Exit criterion: A human reviewer has recorded the design decision and test-plan readiness.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.3 DevSecOps delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall allow media guests to access the press upload service and block their access to the organizer network.; FR-002: The

system shall allow network administrators to revoke media guest access.

- Activity: In an authorized isolated test environment, execute TEST-001: Given: A media guest attempts to access the press upload service When: The media guest submits a valid request to the press upload service Expected: The system shall grant access and return a successful response
- Activity: In an authorized isolated test environment, execute TEST-002: Given: A media guest attempts to access the organizer network When: The media guest submits a request to connect to the organizer network Expected: The system blocks the media guest's access to the organizer network
- Activity: In an authorized isolated test environment, execute TEST-003: Given: A network administrator logs into the administrative interface When: The administrator selects and executes a revocation action for a media guest's access Expected: The system shall process the revocation request and update access status accordingly
- Activity: In an authorized isolated test environment, execute TEST-004: Given: An administrator revokes a media guest's access When: The administrator selects and executes a revocation action for a media guest's access Expected: The media guest is no longer permitted to access the press upload service
- Activity: In an authorized isolated test environment, execute TEST-005: Given: Network administrators have configured media guest access permissions. When: The administrator revokes media guest access for a specific user Expected: The system permits the revoked access to be removed and the user no longer has permission to access media content
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Activity: Review security-check evidence for every delivery change and record a human release decision; keep unresolved tooling and operational choices open.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.
- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.4 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package

- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

Canonical content SHA-256: f9c18d0e1fe8b882dc58374a436041b547ea59302ff43201f300fe041e701fc0